Optimality - What makes one test better than another.

Math 742

2026-03-05

Introduction

Among all statistics whose null distributions we know, which one produces the best decision rule?

Likelihood ratios (LRs) are a fundamental concept in statistics, emerging across diverse areas like hypothesis testing, Bayesian inference, decision theory, information theory, and even machine learning. At its core, the likelihood ratio for two hypotheses H_0 and H_1 is defined as:

$$\Lambda(\theta|\mathbf{x}) = \frac{L(\theta_0|\mathbf{x})}{L(\theta_1|\mathbf{x})}$$

where $\Lambda(\theta|\mathbf{x})$ is the **likelihood function** for the parameter θ given the data $\mathbf{x}$

Likelihood ratios "appear everywhere" because they provide a natural, unified way to quantify evidence in favor of one model or hypothesis over another, grounded in the likelihood principle (which states that all information about parameters comes from the likelihood function).

Examples:

- **Hypothesis Testing (Frequentist Perspective):** In the Neyman-Pearson framework, the likelihood ratio test (LRT) is the cornerstone for constructing tests. The Neyman-Pearson lemma proves that for simple hypotheses (fixed θ_0 vs. fixed θ_1), rejecting H_0 when $\Lambda < k$ (for some threshold k) gives the most powerful test at a fixed significance level α. This extends to composite hypotheses via generalized LRTs, used in ANOVA, regression, and model selection (e.g., comparing nested models like full vs. reduced in logistic regression).
- **Bayesian Inference:** The LR directly links to Bayes' theorem. The posterior odds ratio is the prior odds times the LR (also called the Bayes factor when integrated over parameters). This makes LRs essential for model comparison, evidence accumulation, and updating beliefs—e.g., in sequential analysis or A/B testing.
- **Decision Theory and Information:** LRs minimize expected loss in binary decisions under certain cost structures. In information theory, the log-LR relates to Kullback-Leibler divergence (measuring how much one distribution diverges from another), which underpins concepts like mutual information and entropy. This connects to machine learning, where LRs appear in classifiers (e.g., logistic regression's decision boundary is an LR threshold) and anomaly detection.
- **Practical Ubiquity:** In fields like medicine (diagnostic tests: positive LR = sensitivity / (1-specificity)), forensics (evidence strength), and signal processing (detection theory), LRs provide interpretable ratios of probabilities. They're robust to transformations and often lead to pivotal quantities whose distributions are known under the null, making them computationally tractable.

Here is a brief overview of the likelihood ratio test.

https://www.wallstreetmojo.com/likelihood-ratio-test/

Likelihood Ratio Test

The **likelihood ratio** Λ for testing $H_0 : \theta = \theta_0$ (simple) against a composite alternative $H_1 : \theta \neq \theta_0$ is:

$$\Lambda = \frac{L(\theta_0|\mathbf{x})}{\sup_\theta L(\theta|\mathbf{x})} = \frac{L(\theta_0|\mathbf{x})}{L(\hat{\theta}|\mathbf{x})}$$

where $\hat{\theta}$ is the MLE of θ.

Note: This test will reject H_0 for small values of Λ (i.e. close to 0 indicates a poor fit under H_0 relative to the best fit).

$$-2\log\Lambda = -2\log\left(\frac{L(\theta_0)}{L(\hat{\theta})}\right) = -2\left[\log L(\theta_0) - \log L(\hat{\theta})\right] = 2\left[\ell(\hat{\theta}) - \ell(\theta_0)\right]$$

Note: Multiplying by -2 does the following things:

- makes it positive
- rescales it to match variance geometry
- converts log-likelihood differences into squared standardized distances

Theorem (Wilks' Theorem): $-2\log\Lambda \xrightarrow{d} \chi^2_q$, where q is the difference in the number of free parameters between the full model under H_1 and the restricted model under H_0.

Proof (Intuition / proof sketch):

The key insight comes from a **quadratic approximation** of the log-likelihood around the MLE $\hat{\theta}$:

- Taylor expand $\ell(\theta)$ around $\hat{\theta}$:

$$\ell(\theta) \approx \ell(\hat{\theta}) + \ell'(\hat{\theta})(\theta - \hat{\theta}) + \frac{1}{2}\ell''(\hat{\theta})(\theta - \hat{\theta})^2$$

- Since $\hat{\theta}$ is the MLE, the score function (i.e. first derivative) is zero: $\ell'(\hat{\theta}) = 0$.
- So (middle term vanishes),

$$\ell(\theta) \approx \ell(\hat{\theta}) + \frac{1}{2}\ell''(\hat{\theta})(\theta - \hat{\theta})^2 = \ell(\hat{\theta}) - \frac{1}{2}(\theta - \hat{\theta})^T[-\ell''(\hat{\theta})](\theta - \hat{\theta})$$

Note: The log-likelihood drops off quadratically as you move away from the MLE.

- Multiply by -2, and let $\tilde{\theta}$ be the best fit under the reduced model and $\hat{\theta}$ is the parameter of the best fit under the full model. So, our quadratic approximation becomes:

$$-2\log\Lambda \approx (\tilde{\theta} - \hat{\theta})^T[-\ell''(\hat{\theta})](\tilde{\theta} - \hat{\theta})$$

Interpretation: the LRT statistic is a curvature-weighted squared distance between the unrestricted and restricted fits.

The difference between the unrestricted and restricted estimators lies only in the q constrained directions.

Therefore the quadratic form, as we will see, reduces to the squared length of a $q-$dimensional normal vector.

- Replace the curvature by Fisher information, assuming standard regularity conditions, $-\ell''(\hat{\theta}) \approx nI(\theta_0)$ (Note: The log-likelihood curvature is a sum of iid terms, so by the Law of Large Numbers it converges to its expectation, which is the Fisher information.)

$$-2\log\Lambda \approx n(\tilde{\theta}-\hat{\theta})^T I(\theta_0)(\tilde{\theta}-\hat{\theta})$$

By asymptotic normality of the MLE, we know that $\sqrt{n}(\hat{\theta}-\theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1})$.

Therefore, $\sqrt{n}(\hat{\theta}-\tilde{\theta})$ is asymptotically multivariate normal in the q constrained directions.

Under the null hypothesis, the unrestricted and restricted estimators differ only in the q constrained directions. Thus, In the remaining $p-q$ directions, the estimators agree asymptotically.

- Now, multiply by $I(\theta_0)^{1/2}$ to standardize.
- Define $Z = I(\theta_0)^{1/2}\sqrt{n}(\hat{\theta}-\tilde{\theta})$, so asympotitically $Z \sim N_q(0, I_q)$ so the likelihood ratio becomes:

$$-2\log\Lambda \approx Z^T Z = Z_1^2 + Z_2^2 + ... + Z_q^2$$

And (by definition) this is a χ^2_q random variable.

- The two models differ only in the q constrained directions.
- So the likelihood ratio only measures random movement in those q directions.
- Random movement in q independent normal directions $\implies$ chi-square with q degrees of freedom.

All three classical tests are based on the **same quadratic approximation of the log-likelihood near the true parameter.**

Near the MLE $\hat{\theta}$, the log-likelihood can be approximated by a Taylor Expansion:

$$\ell(\theta) = \ell(\hat{\theta}) - \frac{1}{2}(\theta-\hat{\theta})^T I(\theta_0)(\theta-\hat{\theta})$$

where $I(\theta_0)$ is the Fisher information. The approximation says that near the maximum, the log-likelihood behaves like a downward opening quadratic bowl.

Likelihood Ratio Test (LR)

Compare the log-likelihood at the restricted and unrestricted estimates:

$$-2\log\Lambda = 2[\ell(\hat{\theta}) - \ell(\tilde{\theta})]$$

So the LR statistic measures how much the likelihood drops when we force the parameter to satisfy the null hypothesis.

$\rightarrow$ if the drop is small the parameter is consistent with H_0, if the drop is large, reject H_0.

Wald Test

Measures how far the estimator is from the null value:

$$W = (\hat{\theta}-\theta_0)^T I(\theta_0)(\hat{\theta}-\theta_0)$$

So, Wald looks at the distance from the null value. $\rightarrow$ small distance do not reject H_0, large distance reject H_0.

Score Test (Rao)

Evaluates the slope of the likelihood at the null:

$$S = U(\theta_0)^T I(\theta_0)^{-1} U(\theta_0)$$

Measures how steeply the likelihood is rising at the null.

If we start at H_0 how strongly does the likelihood want to move away? Flat slope $\rightarrow$ do not reject H_0, steep slope reject H_0.

$$\text{All 3 tests converge to } \chi^2_q$$

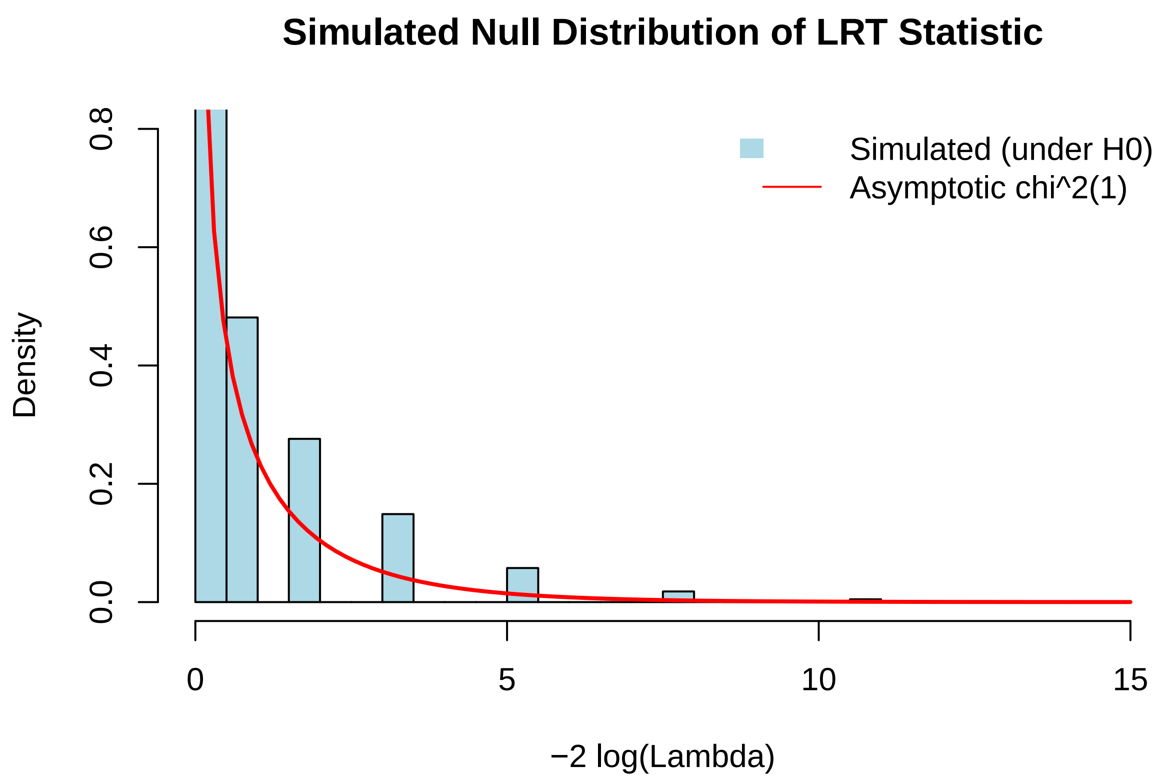
The likelihood surface near the maximum is basically a quadratic bowl, and the LR, Wald, and Score tests are just three different ways of measuring distance on that same bowl.

This R code simulates a likelihood ratio test (LRT) for a binomial proportion:

- We generate data under the null hypothesis $H_0 : p = 0.5$ many times (10,000 simulations).
- For each simulated dataset, compute the LRT statistic $-2 \log \Lambda$, where $\Lambda = L(H_0)/L(H_1)$ and H_1 is the unrestricted MLE.
- Plot the histogram of these statistics to show the empirical null distribution.
- Overlay the asymptotic $\chi^2(1)$ density (Wilk's theorem) for comparison—demonstrating how well the approximation holds even for small $n = 20$. Compute the empirical Type I error rate by checking how often the statistic exceeds the critical value for $\alpha = 0.05$ (should be close to 0.05).

```r
# R Simulation of Likelihood Ratio Test (LRT) for Binomial Proportion
# Frequentist Perspective: Simulate under H0 to show null distribution of -2 log Lambda
# Example: Test H0: p = 0.5 vs H1: p != 0.5 for binomial data (n trials)
# Under H0, -2 log Lambda ~ chi-squared(df=1) asymptotically

# Set parameters
set.seed(123)       # For reproducibility
n <- 20             # Number of trials per sample (small n to show approximation)
p0 <- 0.5           # Null hypothesis value
num_sims <- 10000   # Number of simulations

# Function to compute LRT statistic for one sample
compute_lrt <- function(x, n, p0) {
  # MLE under H1: phat = x/n
  p_hat <- x / n

  # Likelihood under H0: L0 = binom(p0)
  logL0 <- dbinom(x, size = n, prob = p0, log = TRUE)

  # Likelihood under H1: L1 = binom(\hat{p})
  logL1 <- dbinom(x, size = n, prob = p_hat, log = TRUE)

  # LRT statistic: -2 log(L0 / L1)
  test_stat <- -2 * (logL0 - logL1)
  return(test_stat)
}

# Simulate data under H0 and compute test stats
test_stats <- replicate(num_sims, {
  # Generate binomial data under H0
  x <- rbinom(1, size = n, prob = p0)
  compute_lrt(x, n, p0)
```

```r
})

# Plot histogram of simulated test stats
hist(test_stats, breaks = 50, freq = FALSE, col = "lightblue",
     main = "Simulated Null Distribution of LRT Statistic",
     xlab = "-2 log(Lambda)", ylim = c(0, 0.8))

# Overlay asymptotic chi-squared(df=1) density
curve(dchisq(x, df = 1), col = "red", lwd = 2, add = TRUE)

# Add legend
legend("topright", legend = c("Simulated (under H0)", "Asymptotic chi^2(1)"),
       fill = c("lightblue", NA), border = c(NA, NA), lty = c(NA, 1),
       col = c(NA, "red"), bty = "n")
```




```r
# Empirical Type I error rate at alpha=0.05 (critical value from chi^2)
crit_val <- qchisq(0.95, df = 1)  # approx 3.841
type1_error <- mean(test_stats > crit_val)
cat("Empirical Type I error rate (alpha=0.05):", round(type1_error, 4), "\n")
```

```
## Empirical Type I error rate (alpha=0.05): 0.0406
```

Optimality

In the context of LRs, "optimal" typically refers to statistical efficiency or power:

- **Most Powerful Test:** For fixed Type I error (false positive rate α), the LRT maximizes power (1 -

Type II error, or true positive rate) against a specific alternative. This is "uniformly most powerful" (UMP) if it holds for all alternatives in a class (e.g., one-sided tests for exponentials).

- **Minimax or Admissible:** In decision theory, LR-based rules are often minimax (minimizing maximum risk) or admissible (no better rule exists).
- **Asymptotic Optimality:** For large samples, LRTs are efficient under local alternatives and connect to score tests (Rao) or Wald tests, all asymptotically equivalent but with LR often preferred for finite-sample properties.

  For large samples, the LRT is optimal for detecting small departures from the null. The Wald, score, and likelihood ratio tests all use the same quadratic approximation, so asymptotically they give the same answer.

  In finite samples, the LRT is often slightly more stable. Optimality isn't absolute—it's relative to criteria like power, risk, or information, but LRs often achieve it due to their direct use of the data's likelihood.

Note: Optimality assumes correct model specification; misspecification can lead to sub-optimal performance, hence robustness checks (e.g., bootstrap) are key.

Example: Diagnostic Test for Disease

Consider a simple medical diagnostic test for a disease with prevalence 1% ($\mathbb{P}$(Disease)). The test has sensitivity 90% ($\mathbb{P}$(Disease|PT)) and specificity 95% ($\mathbb{P}$(No Disease|NT)).

Likelihood Ratio Calculation:

Test characteristics.

- Sensitivity = Pr(Positive Test | Disease) = 0.90
- Specificity = Pr(Negative Test | No disease) = 0.95
- Disease prevalence = 0.01

Compute likelihood ratios:

$$\mathrm{LR}^+ = \frac{\text{Sensitivity}}{1 - \text{Specificity}} = \frac{0.9}{0.05} = 18$$

A positive test is 18 times more likely in someone with the disease than without it.

$$\mathrm{LR}^- = \frac{1 - \text{Sensitivity}}{\text{Specificity}} = \frac{0.1}{0.95} = 0.105$$

A negative test is about 0.1 times as likely in someone with the disease as in someone without it.

Convert prevalence to odds

$$P(D) = 0.01$$

$$\text{Prior Odds} = \frac{0.01}{0.99} = 0.0101$$

Before testing, there is about 1 diseased person for every 99 healthy people.

Bayesian Inference

Now, update the likelihood ratios (i.e. calculate posterior odds).

$$\text{Posterior Odds} = \text{Prior Odds} \times \mathrm{LR}^+ = 0.0101 \times 18 = 0.1818$$

Now convert odds back to probability,

$$P(D \mid +) = \frac{0.1818}{1 + 0.1818} = 0.1538$$

Even though the test is strong ($\text{LR}^+ = 18$), the disease is rare, so the probability only increases from 1% to about 15%.

If the test is negative:

$$\text{Posterior Odds} = \text{Prior Odds} \times \text{LR}^- = 0.0101 \times .105 = 0.00106 = 0.106\%$$

A negative test reduces the probability from 1% to about 0.1%.

Here's another way to look at it. Imagine we have 10,000 people. 1% → 100 of them have the disease and 9,900 do not.

Because the test sensitivity is 90% percent, 90 people out of the diseased will test positive. And because of specificity, among the healthy, 495 will test positive.

$$P(D \mid +) = \frac{90}{585} = 15.4\%.$$

Likelihood ratios measure strength of evidence. Posterior probability depends on both evidence and base rate.

Optimality:

Likelihood ratios play a central role in both Bayesian and frequentist inference.

- In Bayesian inference, the likelihood ratio updates prior odds to posterior odds.
- In frequentist hypothesis testing, the Neyman–Pearson lemma states that the likelihood ratio test maximizes detection power for a fixed false alarm rate.

In a standard hypothesis test (binary), we reject H_0 if our test statistic is larger than some threshold – call it η.

$$T(\mathbf{x}) > \eta \rightarrow \text{reject } H_0$$

$$T(\mathbf{x}) < \eta \rightarrow \text{do not reject } H_0$$

Probability of Detection

$$P(T(\mathbf{x}) > \eta | H_1) = \int_{\eta}^{\infty} f_{T|H_1}(t)\, dt$$

Probability of False Alarm

$$P(T(\mathbf{x}) > \eta | H_0) = \int_{\eta}^{\infty} f_{T|H_0}(t)\, dt$$

The goal is to design a test statistic $T(\mathbf{x})$ that achieves high probability of detection while keeping the false alarm rate small. When comparing 2 tests, we can use something called a receiver operating curve (ROC).

To make this more concrete, I revisit the diagnostic test for a disease example.

In the diagnostic test example (sensitivity = 90%, specificity = 95%), the test uses a fixed threshold, which corresponds to a single point on the ROC curve:

- False Positive Rate (FPR) $= 1 - \text{specificity} = 0.05$
- True Positive Rate (TPR) $= \text{sensitivity} = 0.90$

A common model assumes overlapping Gaussian distributions for a continuous biomarker:

$$\text{Biomarker} \mid \text{no disease} \sim N(0,1), \qquad \text{Biomarker} \mid \text{disease} \sim N(\mu,1).$$

A diagnostic rule is obtained by choosing a threshold η and classifying a patient as diseased when the biomarker exceeds η.

Different thresholds η produce different pairs of

$$(\text{FPR}, \text{TPR}),$$

where

$$\text{FPR} = P(\text{biomarker} > \eta \mid \text{no disease})$$

and

$$\text{TPR} = P(\text{biomarker} > \eta \mid \text{disease}).$$

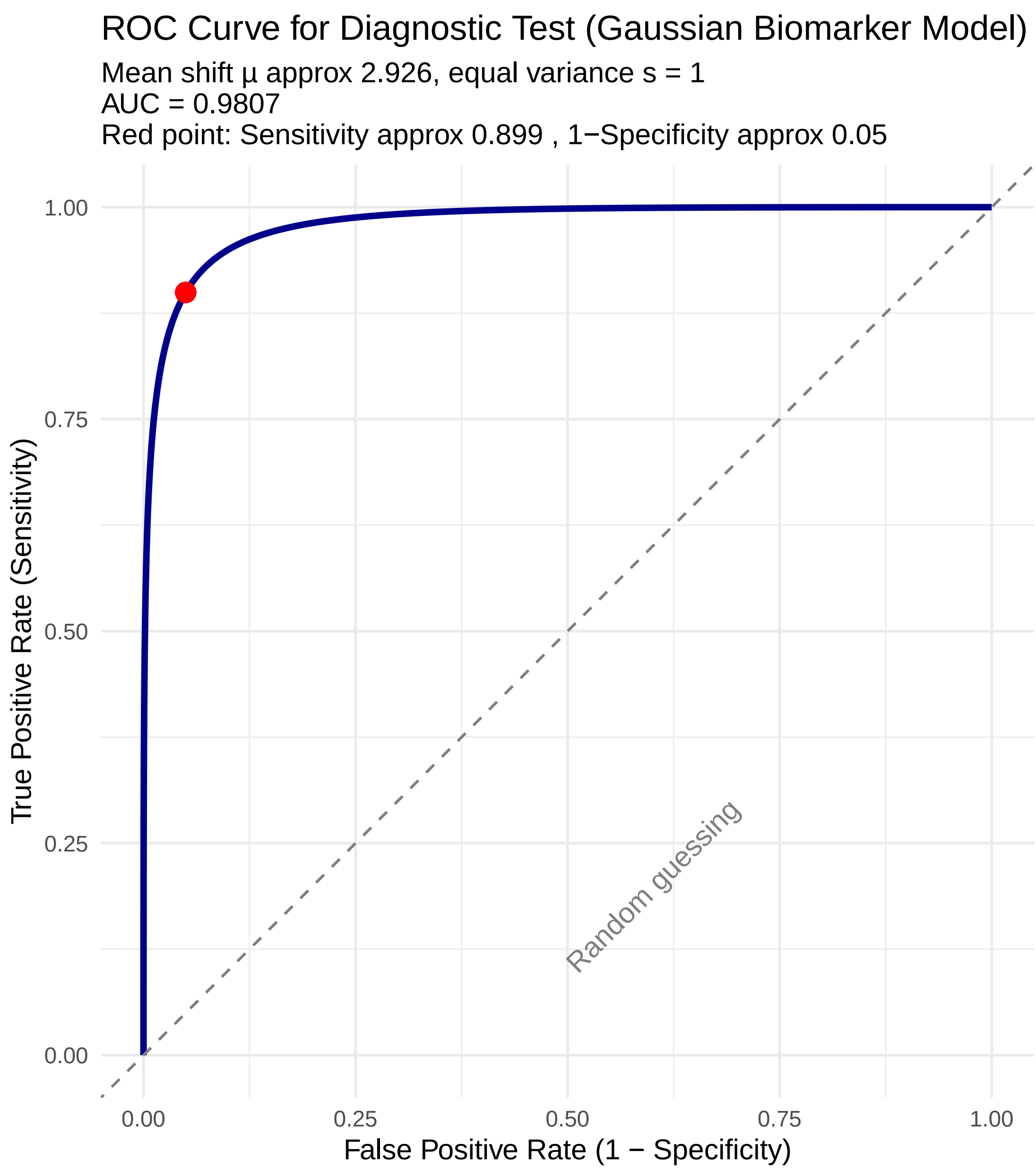
As the threshold η varies, these pairs trace out the ROC curve.

Summary:

The ROC curve plots the true positive rate (TPR) on the y-axis versus the false positive rate (FPR) on the x-axis as the threshold η varies from $-\infty$ to ∞.

The area under the curve (AUC) measures the overall ability of the test to discriminate between diseased and non-diseased individuals:

- $\text{AUC} = 1$: perfect discrimination
- $\text{AUC} = 0.5$: random guessing



- The curve bows strongly toward the top-left, indicating excellent diagnostic performance.
- Theoretical AUC ≈ 0.921 (very good discrimination).
- The red point marks an operating threshold where sensitivity ≈ 0.90 and specificity ≈ 0.952, very close to the original example (90%/95%).

Neyman–Pearson Lemma (Conceptual)

The Neyman-Pearson Lemma,

- Provides theoretical foundation for likelihood ratio tests.
- States likelihood ratio test is the most powerful test.
- Guarantees optimal trade-off between Type I and Type II errors.

- Applies to tests with fixed significance level.
- Applies specifically to simple vs. simple hypotheses (i.e. both H_0 and H_1 state exact parameter values.)

Theorem (Neyman-Pearson): For testing a simple null hypothesis H_0 if the likelihood ratio

$$\Lambda(\mathbf{x}) = \frac{L(\theta_0|\mathbf{x})}{L(\theta_1|\mathbf{x})} \leq k,$$

Reject H_0 when $\Lambda(\mathbf{x}) \leq k$, where the constant k is chosen so that the probability of rejection under H_0 equals α.

Small values of Λ indicate evidence against the null model. Note: You can also write $\Lambda(\mathbf{x})$ as $\frac{L(\theta_1|\mathbf{x})}{L(\theta_0|\mathbf{x})}$. But in that case, you would reject H_0 if $\Lambda(\mathbf{x}) \geq k$.

The test is **optimal**: which means that no other test of size $\leq \alpha$ has higher power under H_1.

Example:

Suppose you want to test whether a coin is "fair". In order to do this you flip the coin one time. You want to test, $H_0 : p = 0.5$ versus $H_1 : p = 0.7$

Likelihood Ratio,

$$\Lambda(x) = \frac{0.5^x(1-0.5)^{1-x}}{0.7^x(0.3)^{1-x}}$$

Compute:

- If $x = 0 : \Lambda(0) = 0.5/0.3 \approx 1.67$
- If $x = 1 : \Lambda(1) = 0.5/0.7 \approx 0.714$

$\Lambda(1) < \Lambda(0)$ so, $x = 1$ is more supportive of H_1

By the Neyman–Pearson Lemma, the most powerful level-α test rejects H_0 when $\Lambda(\mathbf{x}) \leq k_\alpha$

Since $x = 1$ has the smallest likelihood ratio, it provides the strongest evidence against H_0. Because the sample space is discrete, rejecting deterministically would give test size 0.5. To see this,

$$\alpha = \Pr(\text{reject H0 when it's true})$$

Our rejection region is $R = \{1\}$. So we reject when $x = 1$ but if H_0 is true we use $p = 0.5$.

$$\Pr(\text{reject H0 when it's true}) = \Pr(x = 1|p = 0.5) = 0.5$$

.

So, $\alpha = 0.5$.

To obtain a level-0.05 test, we randomize. The rejection rule would be:

- If $X = 1$, reject H_0 with probability 0.1 (i.e. 10% of the time)
- If $X = 0$, never reject.

$$P_{H0}(\text{reject}) = (0.5) \times (0.1) = 0.05$$

Since the sample space has only two points, no non-randomized test achieves size exactly 0.05. Thus the rejection region contains $x = 1$, but we randomize at that point to achieve the desired test size.

Therefore, we reject where the likelihood ratio is smallest, and we randomize only because the sample space is discrete.

In discrete hypothesis testing, the Neyman–Pearson lemma solves a fractional knapsack problem: we spend type-I error probability on outcomes that give the greatest increase in power per unit cost.

Example:

Let $X_1, X_2, ..., X_n$ be an iid sample from $N(\mu, \sigma^2)$, where $\sigma^2 < \infty$ is known.

Perform the following hypothesis test:

$$H_0 : \mu = \mu_0 \text{ versus } H_1 : \mu = \mu_1$$

where $\mu_1 > \mu_0$.

The likelihood function is:

$$L(\mu_i|\mathbf{x}) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{j=1}^{n}(x_j - \mu_i)^2\right), i = 0, 1$$

The likelihood ratio,

$$\Lambda(\mathbf{x}) = \frac{L(\mu_0|\mathbf{x})}{L(\mu_1|\mathbf{x})}$$

Taking logs and simplifying,

$$\begin{aligned}\log\Lambda(\mathbf{x}) &= -\frac{1}{2\sigma^2}\left[\sum_{j=1}^{n}(x_j - \mu_0)^2 - \sum_{j=1}^{n}(x_j - \mu_1)^2\right] \\ &= \frac{1}{2\sigma^2}\left[\sum_{j=1}^{n}(x_j - \mu_1)^2 - \sum_{j=1}^{n}(x_j - \mu_0)^2\right]\end{aligned}$$

Expand the squares,

$$\sum(x_j - \mu)^2 = \sum x_j^2 - 2\mu\sum x_j + n\mu^2$$

So,

$$\begin{aligned}&\sum(x_j - \mu_1)^2 - \sum(x_j - \mu_0)^2 \\ &= -2(\mu_1 - \mu_0)\sum x_j + n(\mu_1^2 - \mu_0^2)\end{aligned}$$

Plug back in,

$$\log\Lambda(\mathbf{x}) = \frac{1}{2\sigma^2}\left(-2(\mu_1 - \mu_0)\sum x_j + n(\mu_1^2 - \mu_0^2)\right)$$

$$= \frac{1}{\sigma^2}(\mu_1 - \mu_0)n\bar{X} - \frac{n(\mu_1^2 - \mu_0^2)}{2\sigma^2}$$

$$= \frac{1}{\sigma^2}(\mu_1 - \mu_0)n\bar{X} - \frac{n(\mu_1 - \mu_0)(\mu_1 + \mu_0)}{2\sigma^2}$$

$$\log \Lambda(\mathbf{x}) = \frac{n}{\sigma^2}(\mu_1 - \mu_0)\left(\bar{X} - \frac{\mu_1 + \mu_0}{2}\right)$$

Rewrite this to highlight the linear structure in $\bar{X}$:

$$\log \Lambda(\mathbf{x}) = \underbrace{\frac{n}{\sigma^2}(\mu_1 - \mu_0)}_{\text{constant}}\,\bar{X} - \underbrace{\frac{n}{\sigma^2}(\mu_1 - \mu_0)\frac{\mu_1 + \mu_0}{2}}_{\text{constant}}$$

Thus,

$$\log \Lambda(\mathbf{x}) = a\bar{X} + b$$

where

$$a = \frac{n}{\sigma^2}(\mu_1 - \mu_0), \qquad b = -\frac{n}{\sigma^2}(\mu_1 - \mu_0)\frac{\mu_1 + \mu_0}{2}.$$

So, when $\mu_1 > \mu_0$ the slope (a) is positive, this means we have an increasing linear function of $\bar{X}$. This means, the larger $\bar{X}$ is, the larger $\log \Lambda(\mathbf{x})$ is.

$\bar{X}$ controls the size of the log likelihood ratio. According to the Neyman-Pearson test, we reject when,

$$\Lambda(x) \leq c$$

or equivalently,

$$\log \Lambda(x) \leq \log c$$

Since $\log \Lambda(x)$ increases with $\bar{X}$ an equivalent way to define the rejection region is to reject when,

$$\bar{X} \geq k$$

for some constant k.

Recall that the likelihood-ratio test rejects when

$$\Lambda(x) \leq c$$

or equivalently,

$$\log \Lambda(x) \leq \log c.$$

The point where the likelihoods are equal occurs when

$$\Lambda(x) = 1$$

or

$$\log \Lambda(x) = 0.$$

$$\log \Lambda(\mathbf{x}) = \underbrace{\frac{n}{\sigma^2}(\mu_1 - \mu_0)}_{\text{constant}} \bar{X} - \underbrace{\frac{n}{\sigma^2}(\mu_1 - \mu_0)\frac{\mu_1 + \mu_0}{2}}_{\text{constant}} = 0$$

The likelihood-ratio test therefore selects the parameter value whose mean is closest to the observed sample mean.

This occurs when

$$\bar{X} = \frac{\mu_1 + \mu_0}{2}$$

which is the midpoint between the two means.

Thus the likelihood-ratio test compares the sample mean to the midpoint between μ_0 and μ_1:

- If $\bar{X}$ is closer to $\mu_0 \to$ evidence favors H_0
- If $\bar{X}$ is closer to $\mu_1 \to$ evidence favors H_1

Limitations

1. The Neyman-Pearson Lemma does not apply to composite hypotheses.

- **Simple:** $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$
- **Composite:** $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$
- Or, depending on what you are trying to show: $H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$
- NP lemma only applies to testing one simple hypothesis against another simple hypothesis.

For composite hypotheses we must use other tools such as:

- Uniformly Most Powerful (UMP) tests (if they exist)
- Generalized likelihood ratio tests (GLRT)

2. Why UMP Tests Rarely Exist.

- UMP = Uniformly Most Powerful test: a test that maximizes power for every value in the alternative hypothesis.
- Why they are rare:

  a) Composite alternatives create conflicts:

$\to$ A test that is most powerful for one alternative parameter value might have low power for another.

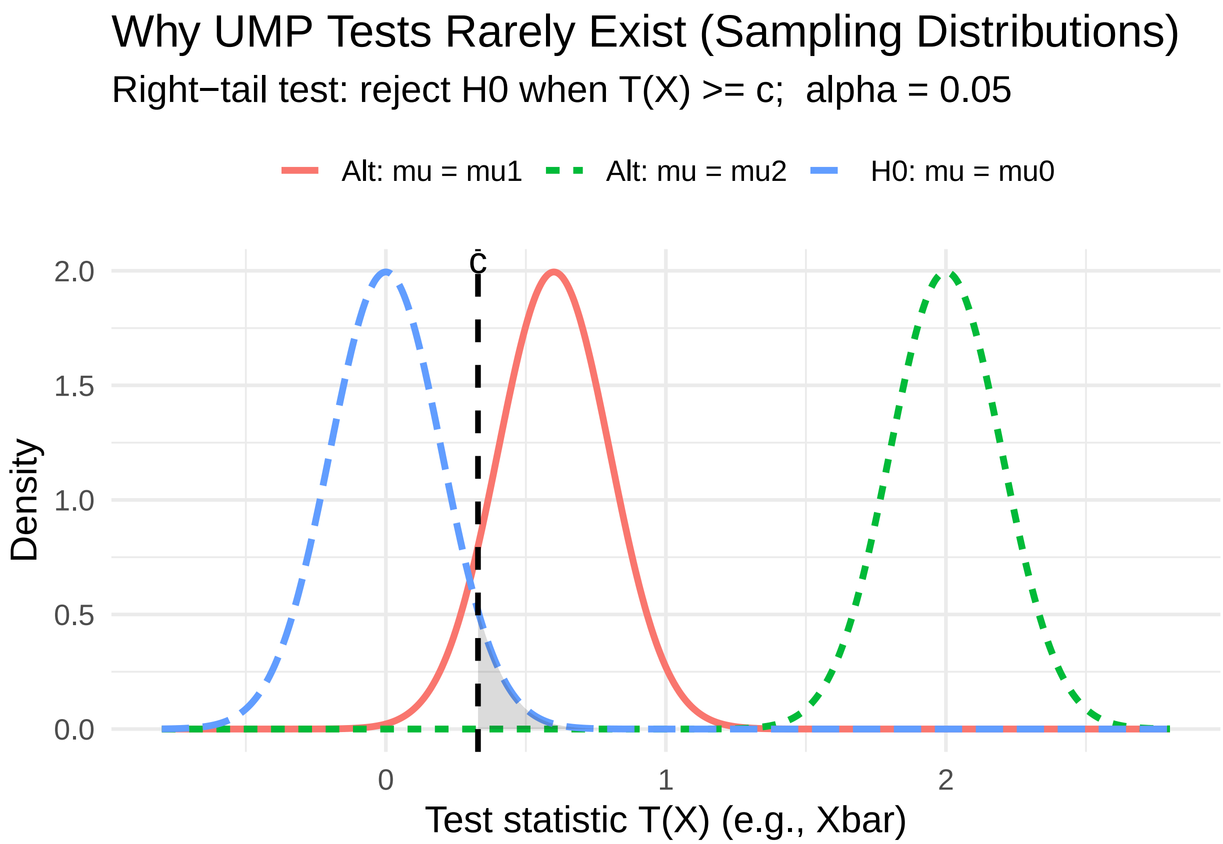
  b) Likelihood ratios may not be monotone:

$\to$ Many important UMP tests exist because the family of distributions has the **monotone likelihood ratio (MLR)** property. The Karlin–Rubin theorem shows that MLR implies the existence of UMP tests for one-sided composite hypotheses.

  c) Higher-dimensional parameters:

$\to$ For multi-parameter problems, even simple one-sided hypotheses rarely allow a UMP test.

Bottom Line: The NP-Lemma gives you the best test for a very specific "point" alternative. A test that is most powerful for one alternative parameter value may have poor power for another value in the alternative set.



Power in these curves would be the area under the curve to the right of the dashed vertical line. For the red curve, this might be 0.85. For the green curve, the area to the right of the dashed vertical line would be 1. The same test produces different power values for different alternatives.

1. The curves show the sampling distribution of a test statistic under different parameter values.
2. The dashed line represents the **null hypothesis parameter value.**
3. Different alternatives (μ_1, μ_2) shift the distribution.
4. A rejection region optimized for detecting μ_1 may not maximize power for μ_2.
5. Therefore no single rejection region is optimal for all alternatives, so a UMP test does not exist.

Conclusion: Since no single rejection rule is best for every value in $\mu > 0$, a uniformly most powerful (UMP) test generally does not exist.

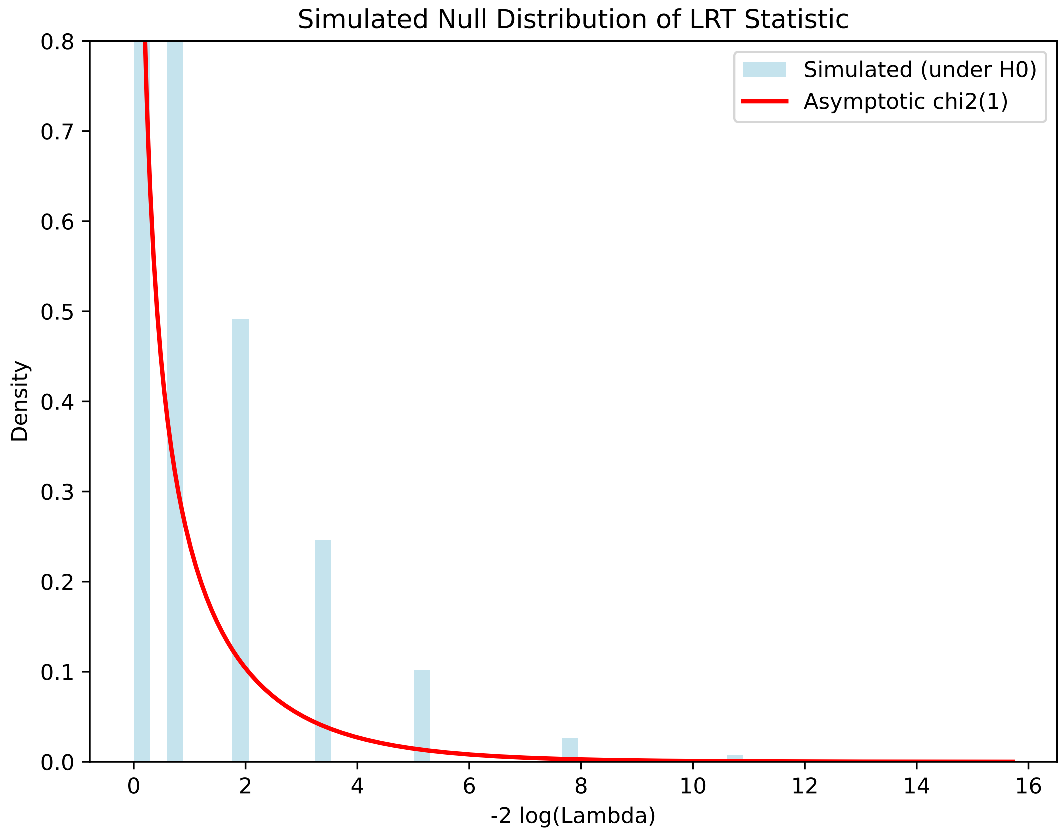
Likelihood ratios are not arbitrary — they arise as the solution to an optimal decision problem.

This is why likelihood ratio tests appear throughout statistics: they emerge naturally when we seek decision rules that maximize power while controlling the Type I error rate.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, chi2
```

```python
# Set parameters (matching the R code)
np.random.seed(123)          # For reproducibility
n = 20                       # Number of trials per sample
p0 = 0.5                     # Null hypothesis value
num_sims = 10000             # Number of simulations

# Function to compute the LRT statistic for one binomial observation x
def compute_lrt(x, n, p0):
    p_hat = x / n
    # Handle boundary cases where p_hat = 0 or 1 (logL1 would be problematic, but LRT=0 when x=0 or x=n
    if x == 0 or x == n:
        return 0.0
    # Log-likelihood under H0
    logL0 = binom.logpmf(x, n, p0)
    # Log-likelihood under H1
    logL1 = binom.logpmf(x, n, p_hat)
    # LRT statistic
    test_stat = -2 * (logL0 - logL1)
    return test_stat

# Simulate data under H0 and compute test statistics
test_stats = np.array([
    compute_lrt(np.random.binomial(n, p0), n, p0)
    for _ in range(num_sims)
])

# Plot histogram of simulated test statistics
plt.figure(figsize=(8, 6))
counts, bins, _ = plt.hist(test_stats, bins=50, density=True, color='lightblue', alpha=0.7,
                           label='Simulated (under H0)')

# Overlay asymptotic chi-squared(df=1) density
# Extend x_grid slightly beyond the data and start from near 0 (avoid x=0 exactly due to inf)
max_stat = test_stats.max()
x_grid = np.linspace(0.01, max_stat + 1, 500)  # Start from small positive to avoid infinity
plt.plot(x_grid, chi2.pdf(x_grid, df=1), color='red', lw=2,
         label='Asymptotic chi2(1)')

# Labels and legend
plt.title("Simulated Null Distribution of LRT Statistic")
plt.xlabel("-2 log(Lambda)")
plt.ylabel("Density")
plt.ylim(0, 0.8)  # Matches original R ylim
plt.legend(loc='upper right')

# Empirical Type I error rate at alpha = 0.05
crit_val = chi2.ppf(0.95, df=1)   # approx 3.841
type1_error = np.mean(test_stats > crit_val)
print(f"Empirical Type I error rate (alpha=0.05): {type1_error:.4f}")

plt.show()
```


Simulated Null Distribution of LRT Statistic
Simulated (under H0)
Asymptotic chi2(1)
Density
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
2
4
6
8
10
12
14
16
-2 log(Lambda)